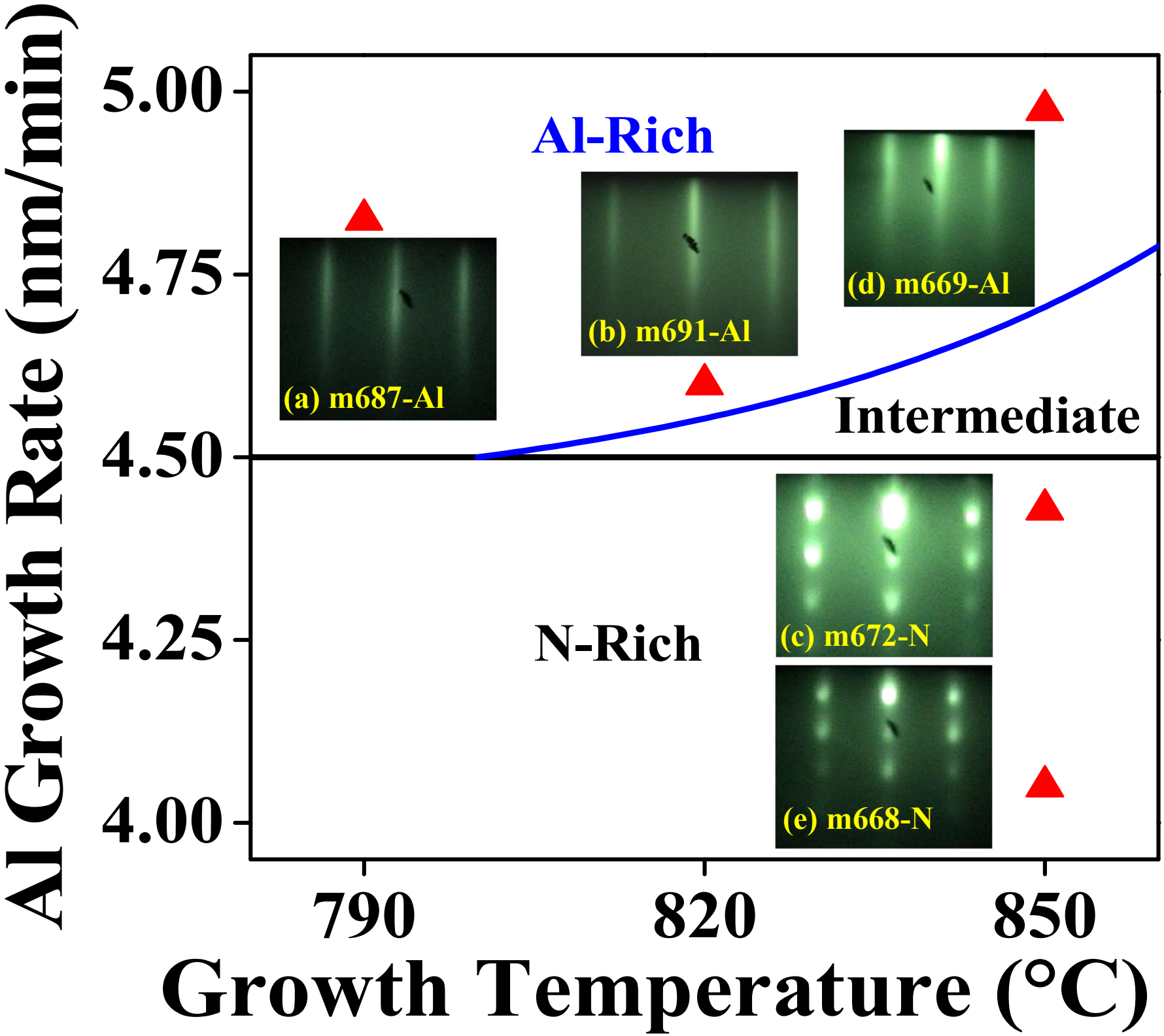
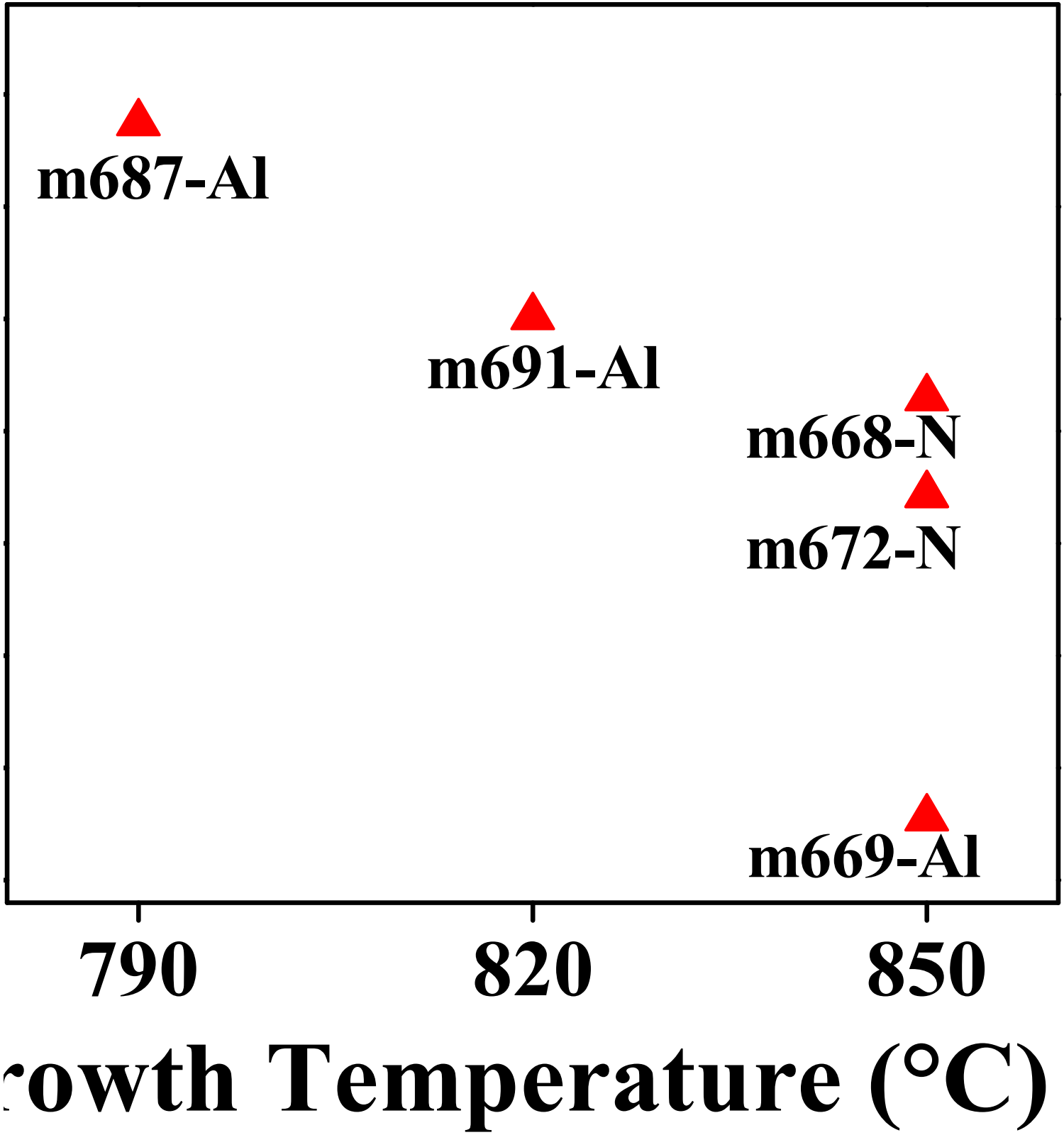
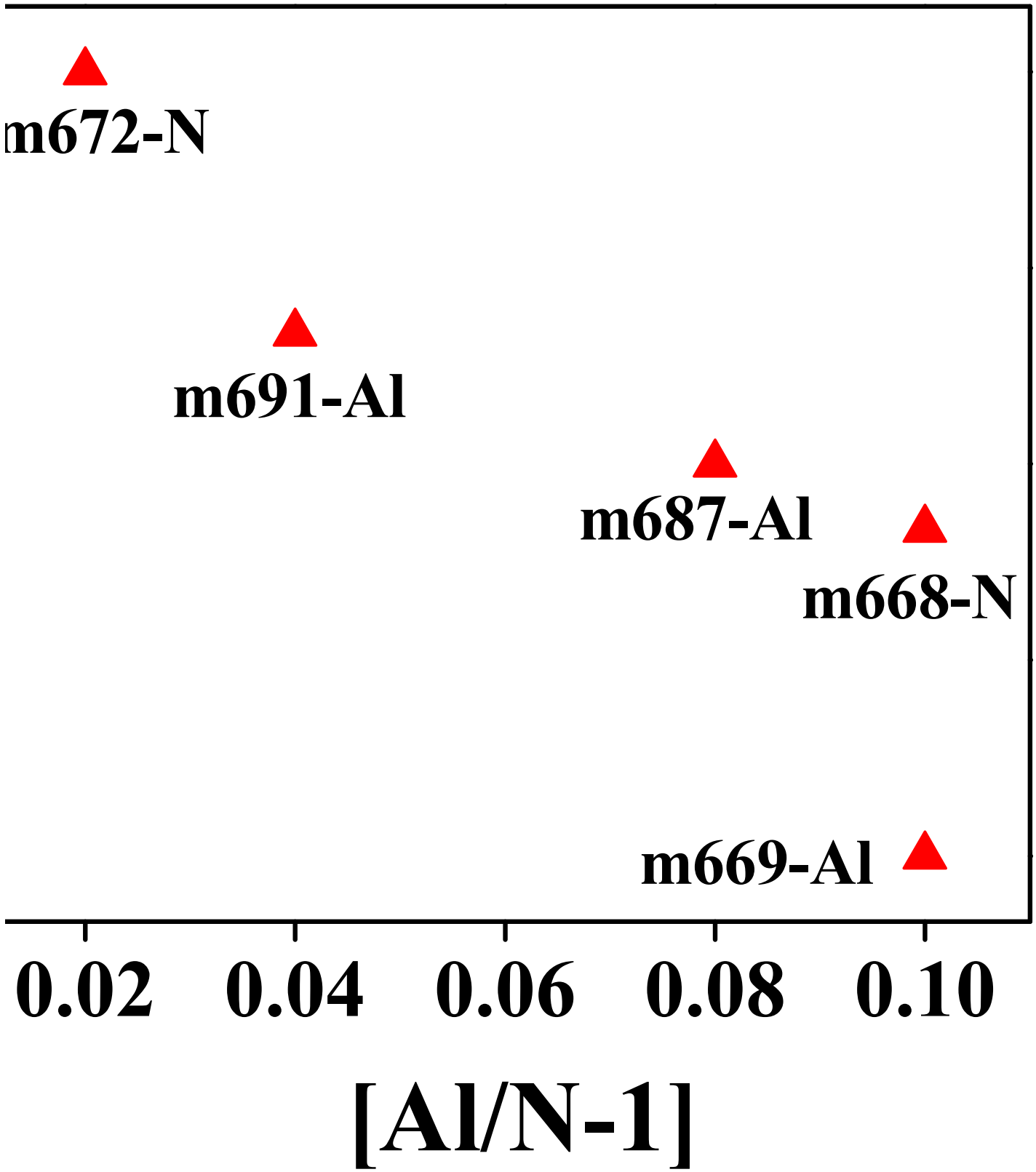
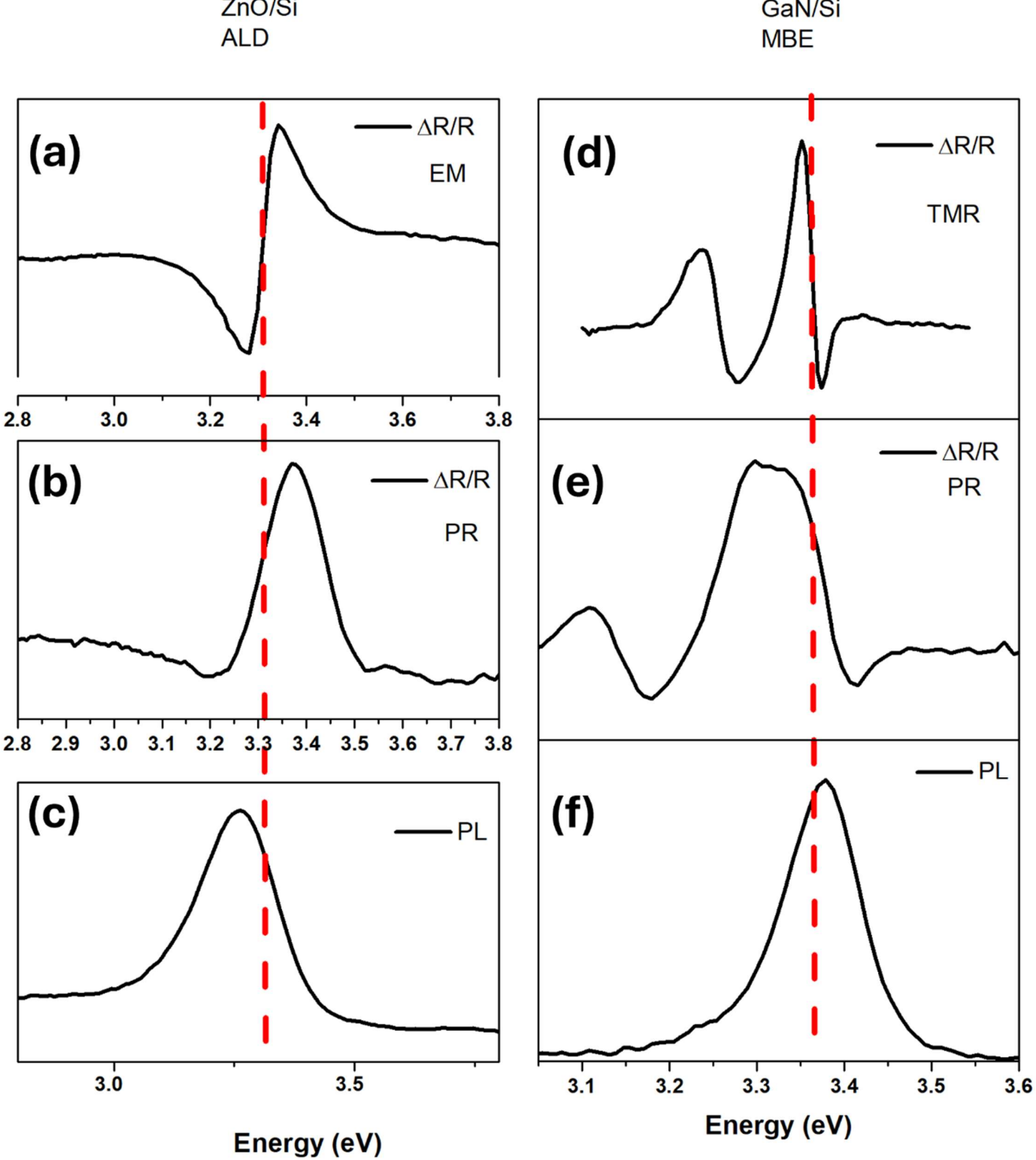

Al Growth Rate (nm/min)
5.00
4.75
4.50
4.25
4.00
Al-Rich
Intermediate
N-Rich
(a) m687-Al
(b) m691-Al
(d) m669-Al
(c) m672-N
(e) m668-N
790
820
850
Growth Temperature (°C)


m687-Al
m691-Al
m668-N
m672-N
m669-Al
790
820
850
rowth Temperature (°C)


m672-N
m691-Al
m687-Al
m668-N
m669-Al
0.02
0.04
0.06
0.08
0.10
[Al/N-1]


ZnO/Si
ALD
GaN/Si
MBE
(a)
ΔR/R
EM
(b)
ΔR/R
PR
(c)
PL
(d)
ΔR/R
TMR
(e)
ΔR/R
PR
(f)
PL
2.8
3.0
3.2
3.4
3.6
3.8
2.9
3.1
3.3
3.5
3.7
Energy (eV)